

Name: _____ Date: _____

Period: _____

Keep in Mind

This is a practice test. It mirrors the real Unit 3 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key.

Part A — Vocabulary (8 pts)

Write the letter of the best description next to each term.

- | | |
|-----------------------------------|--|
| 1. Vertex C | (A) The U-shaped graph of a quadratic function. |
| 2. Axis of symmetry F | (B) The expression $b^2 - 4ac$; its sign tells the number and type of solutions. |
| 3. Zero (root) E | (C) The turning point of a parabola, where it reaches its maximum or minimum. |
| 4. Discriminant B | (D) The number $i = \sqrt{-1}$, which satisfies $i^2 = -1$. |
| 5. Imaginary unit D | (E) An x -value where $f(x) = 0$; an x -intercept of the parabola. |
| 6. Completing the square H | (F) The vertical line $x = h$ that splits a parabola into mirror halves. |
| 7. Parabola A | (G) A quadratic written as $ax^2 + bx + c$. |
| 8. Standard form G | (H) Rewriting $x^2 + bx$ as a perfect-square trinomial by adding $(\frac{b}{2})^2$. |

Part B — Multiple Choice (12 pts)

Circle the best answer.

- The vertex of $y = (x - 4)^2 - 1$ is (A) $(4, -1)$ ← (B) $(-4, -1)$ (C) $(4, 1)$ (D) $(-4, 1)$
- $\sqrt{-25} =$ (A) $5i$ ← (B) -5 (C) $25i$ (D) 5
- The discriminant of $x^2 - 4x + 7 = 0$ is -12 , so the equation has (A) two real solutions (B) one real solution (C) **two complex (non-real) solutions** ← (D) no solutions of any kind
- Factored completely, $x^2 - 49$ is (A) $(x - 7)(x + 7)$ ← (B) $(x - 7)^2$ (C) $(x + 7)^2$ (D) prime
- The axis of symmetry of $y = x^2 - 6x + 5$ is (A) $x = 3$ ← (B) $x = -3$ (C) $x = 6$ (D) $x = 5$
- A line and a parabola can intersect in *at most* how many points? (A) 1 (B) **2** ← (C) 3 (D) infinitely many

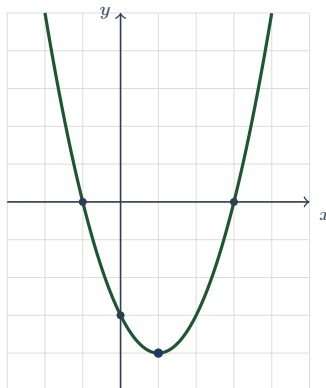
Teacher Note

Item 2: pull the i out first — $\sqrt{-25} = \sqrt{25} \sqrt{-1} = 5i$ (choice C forgets to simplify $\sqrt{25}$). Item 3: $b^2 - 4ac < 0 \Rightarrow$ the $\pm\sqrt{}$ is imaginary, so two complex conjugate roots and *zero* x -intercepts (D is wrong — there *are* solutions, just not real). Item 5: $x = -\frac{b}{2a} = \frac{6}{2} = 3$.

Part C — Short Answer & Computation (40 pts)

Show your work.

1. **Read the graph.** The parabola f is shown below.



Vertex: $(1, -4)$; axis of symmetry: $x = 1$; y -intercept: $(0, -3)$;
zeros: $x = -1$, $x = 3$; minimum or maximum value: **minimum** -4 ; range: $y \geq -4$

2. **From standard form.** For $f(x) = x^2 - 6x + 5$, use $x = -\frac{b}{2a}$ to find the vertex, give the axis of symmetry, and state whether the parabola opens up or down. $x = -\frac{-6}{2(1)} = 3$; $f(3) = 9 - 18 + 5 = -4$, **so the vertex is $(3, -4)$. Axis of symmetry $x = 3$. Since $a = 1 > 0$, it opens up (so -4 is a minimum).**

3. **Solve by factoring.**

$$x^2 + 2x - 15 = 0$$

$$(x + 5)(x - 3) = 0 \Rightarrow x = -5 \text{ or } x = 3. \text{ Check: } (-5)^2 + 2(-5) - 15 = 0 \checkmark, 3^2 + 2(3) - 15 = 0 \checkmark.$$

4. **Solve using square roots.**

$$(x - 2)^2 = 49$$

$$x - 2 = \pm 7 \Rightarrow x = 2 + 7 = 9 \text{ or } x = 2 - 7 = -5.$$

5. **Solve by completing the square.**

$$x^2 + 6x - 7 = 0$$

$$x^2 + 6x = 7; \text{ add } \left(\frac{6}{2}\right)^2 = 9 \text{ to both sides: } (x + 3)^2 = 16; x + 3 = \pm 4, \text{ so } x = 1 \text{ or } x = -7.$$

6. **Complex numbers.** Let $z_1 = 3 + 2i$ and $z_2 = 1 - 4i$. Write each answer in $a + bi$ form.

$$(a) \ z_1 + z_2 = 4 - 2i \quad (b) \ z_1 - z_2 = 2 + 6i$$

$$(c) \ z_1 \cdot z_2 = (3 + 2i)(1 - 4i) = 3 - 12i + 2i - 8i^2 = 3 - 10i + 8 = 11 - 10i$$

7. **Quadratic formula & discriminant.** Solve $2x^2 + 3x - 2 = 0$. First compute the discriminant and state how many real solutions to expect, then solve. $b^2 - 4ac = 3^2 - 4(2)(-2) = 9 + 16 = 25 > 0$, **so two real solutions.**

$$x = \frac{-3 \pm \sqrt{25}}{2(2)} = \frac{-3 \pm 5}{4}, \text{ giving } x = \frac{1}{2} \text{ or } x = -2.$$

8. **System.** Solve the linear-quadratic system algebraically. Give every solution as an ordered pair.

$$y = x^2 - 2x - 3 \quad y = x - 3$$

Set equal: $x^2 - 2x - 3 = x - 3 \Rightarrow x^2 - 3x = 0 \Rightarrow x(x - 3) = 0$, **so $x = 0$ or $x = 3$. Then $y = x - 3$:**
 $x = 0 \Rightarrow (0, -3)$; $x = 3 \Rightarrow (3, 0)$. **Two solutions: $(0, -3)$ and $(3, 0)$.**

Teacher Note

Item 1: accept “opens up / minimum” language; range is $y \geq -4$ (closed, vertex included). Item 6c: full credit requires distributing all four terms *and* replacing $i^2 = -1$ — watch for students who leave $-8i^2$. Item 7: since the discriminant is a perfect square (25), the roots are rational; a non-perfect-square discriminant would leave a radical.

Part D — Extended Response (12 pts)

Explain your reasoning in full sentences.

1. **Projectile.** A ball is thrown upward from a rooftop. Its height (in feet) after t seconds is

$$h(t) = -16t^2 + 32t + 48.$$

(a) From which feature of the model do you read the ball’s *starting height*? Give that height. (b) Find the *maximum* height and the time it occurs. Which feature of the graph is this? (c) When does the ball hit the ground? Which feature answers this, and why do you reject the other solution? **(a) The starting height is the y -intercept, $h(0) = 48$ feet. (b) The maximum is the *vertex*. $t = -\frac{b}{2a} = -\frac{32}{2(-16)} = 1$ second; $h(1) = -16 + 32 + 48 = 64$ feet. So the ball reaches a maximum height of 64 ft at $t = 1$ s. (c) The ground is $h = 0$ — the *zeros*. $-16t^2 + 32t + 48 = 0 \Rightarrow t^2 - 2t - 3 = 0 \Rightarrow (t-3)(t+1) = 0$, so $t = 3$ or $t = -1$. We reject $t = -1$ because a negative time is meaningless here; the ball lands at $t = 3$ seconds.**

2. **Discriminant reasoning.** Consider the equation $x^2 - 4x + k = 0$, where k is a constant. (a) Write the discriminant in terms of k . (b) For what values of k does the equation have two real solutions? one? two complex solutions? (c) Explain how the discriminant is connected to the number of x -intercepts of the parabola $y = x^2 - 4x + k$. **(a) $b^2 - 4ac = (-4)^2 - 4(1)(k) = 16 - 4k$. (b) Two real: $16 - 4k > 0 \Rightarrow k < 4$. One (double) real: $16 - 4k = 0 \Rightarrow k = 4$. Two complex: $16 - 4k < 0 \Rightarrow k > 4$. (c) The real solutions are exactly the x -values where the parabola meets the x -axis. So a positive discriminant ($k < 4$) means two x -intercepts, a zero discriminant ($k = 4$) means one (the vertex sits on the x -axis), and a negative discriminant ($k > 4$) means no x -intercepts (the whole parabola stays above the axis).**

Teacher Note

Scoring Part D (6 pts each). D1: 2 pts (a) y -intercept named *and* 48 ft; 2 pts (b) vertex $t = 1$, height 64 ft; 2 pts (c) zeros, $t = 3$, with the $t = -1$ rejection reasoned from context. D2: 1 pt (a) $16 - 4k$; 3 pts (b) all three cases with correct boundary $k = 4$; 2 pts (c) ties discriminant sign to the *count of x -intercepts*. Do not give full (c) credit for restating (b) without the graph connection.